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Phonon-Assisted Topological Stress-Defect Annihilation in Laser-Drilled Glass

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Abstract— *We present a theoretical framework for phonon-assisted topological stress-defect annihilation in laser-drilled glass substrates. By formal analogy to nematic liquid crystal defect dynamics, we model residual stress concentrations around Bessel beam laser-drilled through-glass vias (TGVs) as topological stress-defects characterized by an order parameter S_σ. Acoustic phonon scattering at stress-defect cores mediates viscous relaxation under selective laser annealing. The phonon-mediated annihilation rate couples the Debye frequency, activation energy, and stress order parameter. Three validation experiments are proposed: in-situ Mueller matrix polarimetry, Brillouin light scattering, and micro-Raman spectroscopy. For borosilicate glass with 100-μm via pitch at $0.95T_g$, the predicted annihilation time is ~0.5 s. This framework advances stress management from empirical annealing to predictive, phonon-informed engineering for HBM glass substrates, foldable display UTG/HTG, and optical components.*

Index Terms*—Topological defect, phonon, stress birefringence, laser drilling, through-glass via, Mueller matrix, Brillouin scattering, HBM packaging, glass substrate.*

I. INTRODUCTION

THE demand for high-bandwidth memory (HBM) and advanced heterogeneous integration has driven rapid development of glass core substrates as alternatives to organic interposers [1], [2]. Glass offers superior dimensional stability, lower coefficient of thermal expansion (CTE) mismatch with silicon, and excellent electrical properties at high frequencies [3]. Through-glass vias (TGVs), typically fabricated by Bessel beam or ultrafast laser drilling, provide the critical vertical interconnects in these substrates [4].

However, laser drilling inevitably introduces residual stress fields around each via. These stress concentrations, arising from rapid thermal cycling and material ablation, create spatially distributed birefringence patterns measurable by polarimetric methods [5], [6]. In the context of HBM packaging, where thousands of fine-pitch TGVs must be fabricated at scale [2], residual stress poses a fundamental yield limitation: stress concentrations serve as crack initiation sites during subsequent thermal processing steps [7].

Current approaches to stress management in laser-drilled glass rely on empirical bulk annealing schedules—furnace treatment at temperatures near the glass transition temperature T_g for extended durations [8]. This approach is spatially non-selective, time-consuming, and lacks predictive capability for determining optimal process parameters.

In a parallel domain, the physics of topological defects in nematic liquid crystals provides a rich mathematical framework for understanding how singularities in ordered fields attract, interact, and annihilate [9]–[11]. Chono and Tsuji [12] demonstrated through numerical simulation that paired defects of opposite topological charge undergo mutual annihilation, releasing elastic energy and inducing flow. The Doi theory with Marrucci-Greco potential [13], [14] captures both shortrange and long-range order effects during this process.

In this paper, we establish a formal analogy between LC defect annihilation and glass stress relaxation, and extend it by incorporating phonon dynamics as the microscopic transport mechanism. We introduce a stress order parameter S_σ directly measurable by Mueller matrix polarimetry [15], derive the phonon-mediated annihilation rate, and propose three complementary experiments for validation. This framework transforms stress characterization from a purely diagnostic technique into an active stress-engineering methodology.

II. THEORETICAL FRAMEWORK

A. Stress Order Parameter

In nematic liquid crystals, the degree of molecular alignment is quantified by the scalar order parameter S, derived from the traceless symmetric tensor $\mathbf{S} = \langle \mathbf{uu} - \delta/3 \rangle$ [13]:

$$S = \sqrt{\frac{3}{2}\boldsymbol{S} : \boldsymbol{S}} \quad (1)$$

By analogy, we define the stress order parameter for a glass stress field:

$$S_\sigma(\mathbf{r}) = \frac{1}{\sigma_0}\sqrt{\frac{3}{2}\,\boldsymbol{\sigma}_{dev} : \boldsymbol{\sigma}_{dev}} \quad (2)$$

where the deviatoric stress tensor is

$$\boldsymbol{\sigma}_{dev} = \boldsymbol{\sigma} - \tfrac{1}{3}\mathrm{tr}(\boldsymbol{\sigma})\,\boldsymbol{\delta} \quad (3)$$

σ_0 is the maximum principal stress difference at the via wall, and δ is the identity tensor. At a stress-free point, $S_\sigma = 0$; at maximum concentration, $S_\sigma \to 1$. Crucially, S_σ maps directly to measurable birefringence via the photoelastic law [5]:

$$\Delta n(\mathbf{r}) = C \cdot (\sigma_1 - \sigma_2) = C \cdot \sigma_0 \cdot S_\sigma(\mathbf{r}) \quad (4)$$

where C is the stress-optical coefficient (Brewster's constant, typically 2.5–3.5 nm/cm/MPa for borosilicate glass [16]).

B. Topological Structure of Stress Fields

In nematic LCs, the elastic free energy is given by the Frank-Oseen expression [17]:

$$F = \frac{1}{2}\int \left[K_1(\nabla\cdot\mathbf{n})^2 + K_3(\mathbf{n}\times\nabla\times\mathbf{n})^2\right]dV \quad (5)$$

Point defects (disclinations) arise where the director field **n** is singular, characterized by topological charge s = ±1/2. Analogously, the principal stress direction field $\mathbf{e}_\sigma(\mathbf{r})$ around a laser-drilled via exhibits a topological singularity with winding number:

$$q = \frac{1}{2\pi}\oint_C d\theta_\sigma \quad (6)$$

For an isolated via in an infinite plate under thermoelastic loading, the stress field is radially symmetric with q = +1 [18]. When two adjacent vias have opposing dominant stress characters (compressive vs. tensile), they form a topological defect pair with net charge zero—the precise condition for mutual annihilation in LC theory [10], [12].

C. Phonon-Mediated Stress Relaxation

While the macroscopic analogy provides the topological framework, phonons supply the microscopic mechanism. In stressed glass, the local acoustic phonon dispersion is modified [19]:

$$\omega_L(\mathbf{r}) = \omega_{L,0}\sqrt{1 + \frac{\sigma_{11}(\mathbf{r})}{K + \frac{4}{3}G}} \quad (7)$$

where ω_L is the longitudinal acoustic phonon frequency, K is the bulk modulus, and G is the shear modulus. Compressive stress increases ω_L; tensile stress decreases it. This shift is directly measurable by Brillouin light scattering (BLS) [20], providing an independent stress probe complementary to birefringence. The fractional velocity shift is:

$$\frac{\Delta v_L}{v_{L,0}} = \frac{\sigma_{11}}{2(K + \frac{4}{3}G)} \quad (8)$$

Near T_g, the enhanced phonon population drives viscous relaxation of the Si–O network. The phonon-mediated relaxation rate is:

$$\Gamma_{ph}(T, S_\sigma) = \nu_D \exp\left(-\frac{E_a}{k_B T}\right)\cdot\left[1 - S_\sigma^2\left(\frac{T}{T_g}\right)\right] \quad (9)$$

where ν_D is the Debye cutoff frequency (~10^{13} Hz for borosilicate glass [21]), E_a is the activation energy for Si–O bond rotation (~2–4 eV [22]), and the factor $[1 - S_\sigma^2(T/T_g)]$ captures the self-limiting nature of the process: as stress relaxes ($S_\sigma \to 0$), the thermodynamic driving force vanishes.

D. Annihilation Dynamics

Combining the topological framework with phonon kinetics, the governing equation for S_σ evolution during selective annealing is:

$$\frac{\partial S_\sigma}{\partial t} = D_{th}\nabla^2 S_\sigma - \Gamma_{ph}(T, S_\sigma) \cdot S_\sigma + \frac{F_{el}}{l^2\eta(T)} \quad (10)$$

The three terms represent: (i) thermal diffusion of stress, analogous to elastic diffusion in the Doi theory [13]; (ii) phonon-mediated local relaxation; and (iii) elastic interaction between defect pairs separated by distance l, with η(T) the temperature-dependent viscosity and F_{el} the elastic interaction energy. The characteristic annihilation time is:

$$t_a = \frac{l^2}{D_{th}} \cdot \frac{\eta(T)}{K} \cdot \exp\left(\frac{E_a}{k_B T}\right) \quad (11)$$

For borosilicate glass ($T_g \approx 820$ K, $D_{th} \approx 5\times10^{-7}$ m^2/s, K ≈ 35 GPa) with inter-via spacing l = 100 μm and annealing at $0.95T_g$: $t_a \approx 0.5$ s—well within practical process windows for inline manufacturing.

E. Post-Annihilation Residual State

For annealing time $t \gg t_a$, the residual birefringence decays as:

$$\Delta n_{res}(t) = C\sigma_0 S_{\sigma,0}\ \exp\left(-\frac{t}{\tau_{ph}}\right) \quad (12)$$

where $\tau_{ph}^{-1} = \Gamma_{ph}(T_{anneal})$. For 99% stress reduction: $t_{process} \geq 4.6\tau_{ph}$. Additionally, micro-Raman spectroscopy can independently verify structural relaxation through the Grüneisen relation [23]:

$$\Delta\nu_R \approx \gamma_G \cdot \sigma_{hydro} \quad (13)$$

where γ_G is the mode Grüneisen parameter and σ_{hydro} is the hydrostatic stress component.

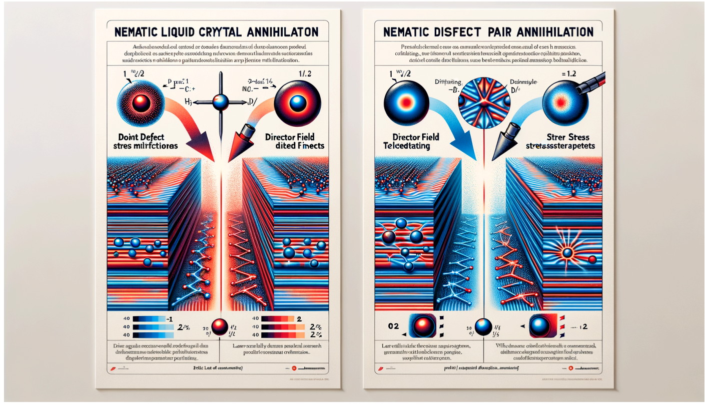

Fig. 1. Conceptual analogy. Left: Nematic LC defect pair (+1/2 and −1/2 disclinations) attract and annihilate, restoring uniform director field. Right: Compressive and tensile stress concentrations around adjacent laser-drilled TGVs undergo analogous mutual annihilation under selective CO_2 laser annealing near T_g.

III. FORMAL ANALOGY MAPPING

Table I summarizes the term-by-term correspondence between the nematic LC defect system and the glass stress-defect system, with the phonon bridge connecting microscopic mechanisms.

TABLE I: FORMAL ANALOGY BETWEEN LC AND GLASS SYSTEMS

Nematic LC System	Glass Stress System	Phonon Bridge
Director $\mathbf{n}(\mathbf{r})$	Principal stress direction $\mathbf{e}_\sigma$	Phonon polarization follows $\mathbf{e}_\sigma$
Order parameter S	Stress order parameter S_σ	Phonon velocity anisotropy $\propto S_\sigma$
Topological charge s = ±1/2	Stress singularity index q = ±1	Scattering cross-section → ∞ at core
Frank elastic constants K_i	Elastic stiffness tensor C_{ij}	Both set interaction range
Rotary diffusivity $\bar{D}$	Thermal diffusivity D_{th}	Phonon MFP $\Lambda \sim D_{th}/v_s$
Maier-Saupe potential U	Elastic strain energy W_{el}	Phonon DOS modified by stress
Thermal fluctuation k_BT	Laser annealing energy	Bose-Einstein $\bar{n}_{BE}(T)$ drives relaxation

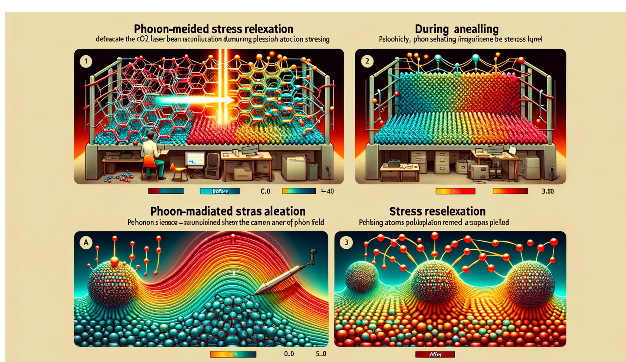

Fig. 2. Phonon-mediated stress relaxation mechanism. (a) Before annealing: distorted Si–O–Si network scatters acoustic phonons, creating localized softening. (b) During CO_2 laser annealing: enhanced phonon population near T_g drives viscous flow and bond-angle relaxation. (c) After annealing: restored equilibrium network with uniform phonon propagation and $S_\sigma \to 0$.

IV. PROPOSED EXPERIMENTS

A. In-situ Mueller Matrix Polarimetry

OBJECTIVE: VALIDATE EQ. (10) — $S_\Sigma(R,T)$ DYNAMICS

Setup: AxoStep/AxoScan polarimeter + CO_2 laser (10.6 μm, 1–10 W) + IR-transparent viewport + Bessel-drilled borosilicate glass (Schott D263, TGV pitch 100–200 μm).

Protocol: Map $\Delta n(\mathbf{r})$ before annealing. Apply selective laser heating at 0.85, 0.90, 0.95 T_g. Re-map at intervals. Fit $S_\sigma(t)$ decay to extract $\Gamma_{ph}(T)$. Construct Arrhenius plot: $\ln(\Gamma_{ph})$ vs. $1/T$ yields E_a.

Expected result: Exponential S_σ decay with $E_a \approx$ 2–4 eV, consistent with Si–O network relaxation [22].

B. Brillouin Light Scattering

OBJECTIVE: INDEPENDENT PHONON-BASED VERIFICATION VIA EQ. (8)

Setup: Micro-Brillouin spectrometer (Sandercock tandem Fabry-Pérot, ~5 μm spatial resolution) [20], [24].

Protocol: Line scan across TGV pair (−200 to +200 μm). Measure longitudinal acoustic phonon velocity $v_L = \omega_L \lambda/2$ before and after annealing.

Expected result: $\Delta v_L/v_0 \approx$ 1–3% at stress concentration [25]. After annealing: v_L returns to bulk value. Spatial $\Delta v_L(\mathbf{r})$ should mirror $\Delta n(\mathbf{r})$.

Novelty: First demonstration of BLS tracking stress-defect annihilation dynamics.

C. Micro-Raman Spectroscopy

OBJECTIVE: MOLECULAR CONFIRMATION VIA EQ. (13)

Setup: Confocal Raman microscope (532 nm excitation, ~1 μm spot) mapping 440 cm^{-1} (Si–O–Si bending) and 800 cm^{-1} (Si–O stretching) bands [23], [26].

Protocol: 2D map around TGV pair. Stress → blue-shift of 440 cm^{-1}. After annealing: peak returns to equilibrium.

Expected result: $\Delta\nu_R \propto S_\sigma$ with sensitivity ~1 cm^{-1}/GPa [26].

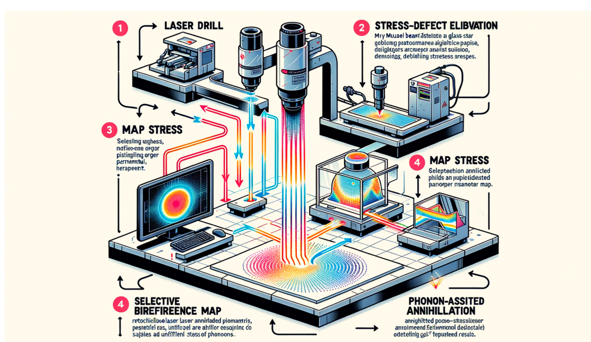

Fig. 3. Experimental workflow. Step 1: Bessel beam laser drilling. Step 2: Mueller matrix polarimetry maps $S_\sigma(\mathbf{r})$. Step 3: Selective CO_2 annealing at identified stress hotspots (phonon-assisted relaxation). Step 4: Multi-modal verification: polarimetry (Δn), Brillouin (Δv_L), Raman ($\Delta\nu_R$).

V. DISCUSSION

Advantages over empirical annealing: The phonon-mediated framework provides (i) a predictive model for t_a at any temperature via Eq. (11), eliminating trial-and-error; (ii) spatial selectivity—only stress hotspots identified by polarimetry require treatment, reducing thermal budget; (iii) three independent cross-validation measurables (birefringence, phonon velocity, Raman shift).

Limitations and scope: The topological analogy is approximate—glass lacks true long-range orientational order. The winding number description is valid when $l \gg r_{via}$. Non-linear viscoelastic effects near T_g may cause deviations from Eq. (10) at $S_\sigma > 0.8$. Extension to the full viscoelastic constitutive model [27] would improve accuracy in the strongly nonlinear regime.

Application scope: Beyond HBM glass substrates, this framework applies to (i) foldable display UTG/HTG—pre-emptive stress mitigation at fold zones to prevent cracking [28]; (ii) glass interposers for chiplet architectures [3]; (iii) architectural safety glass—tempered glass stress distribution optimization [16]; and (iv) photonic waveguide fabrication where birefringence must be controlled to sub-nm levels [29].

VI. CONCLUSION

We have established a phonon-assisted topological framework for understanding and controlling residual stress in laser-drilled glass. The key contributions are: (1) a formal analogy between LC defect annihilation and glass stress relaxation grounded in topological mathematics; (2) S_σ as a measurable order parameter linking theory to existing Mueller matrix technology; (3) phonon dynamics as the microscopic relaxation mechanism with three experimentally verifiable predictions; and (4) a paradigm shift from "measure and accept" to "measure, predict, and cure" in glass stress engineering.

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